RC Aircraft Airfoil Selection and Aerodynamic Analysis Technical Report

1. Summary & Mission Profiles

This technical report provides a design trade study and aerodynamic evaluation for two distinct radio-controlled (RC) aircraft configurations:

- **Aircraft A (Maximum Payload/Heavy Lift):** Optimized to maximize total payload capacity while retaining stable, controllable flight at low cruise speeds.
- **Aircraft B (Maneuverability/Aerobatics):** Optimized for high agility, fast roll rates, symmetrical inverted/upright flight capabilities, and sharp pitch response across high angle-of-attack maneuvers.

2. Airfoil Shortlisting & Mission Candidates

Candidate Profiles for Aircraft A (Payload / Heavy Lift)

- **Selig S1210 (Selected):** High-camber (6.7%), moderate thickness (12.0%) profile designed specifically for micro-heavy lift competitions (e.g., SAE Aero Design). Maximizes vertical lift at low flight speeds while maintaining predictable control.
- **Wortmann FX 63-137:** High-lift glider/payload profile with 5.94% camber and 13.59% thickness. Combines high aerodynamic efficiency (L/D = 89) with high structural thickness for internal spars.
- **Clark-Y:** Classic flat-bottom baseline profile (3.55% camber, 11.72% thickness). Known for forgiving stall behavior, simple construction, and moderate lift capacity.

Candidate Profiles for Aircraft B (Maneuverability / Aerobatics)

- **NACA 0012 (Selected):** Standard fully symmetrical (0.0% camber, 12.0% thickness) aerobatic benchmark. Identical lift characteristics in upright and inverted flight, zero pitching moment (Cm = 0), and predictable trailing-edge stall.
- **NACA 0009:** Thin symmetrical profile (0.0% camber, 9.0% thickness). Low zero-lift drag (Cd,min = 0.0044) and reduced boundary layer displacement yield rapid roll rates, though with a sharper leading-edge stall onset.
- **NACA 23012:** High-lift 5-digit profile (1.84% camber, 12.0% thickness) with near-zero pitching moment (Cm,c/4 ≈ 0), providing strong lift with reduced trim drag.

3. Airfoil Selection & Justification

Aircraft A Selection: Selig S1210

- **Maximum Payload Capacity:** The S1210 achieves a high maximum lift coefficient (CL,max = 1.91), significantly outperforming baseline profiles like the Clark-Y (1.40) and

FX 63-137 (1.66). This directly reduces stall speed and enables carrying higher takeoff weight.
- **Camber Behavior:** Features 6.7% camber located at 51.1% chord. The high camber acts like a permanent flap deployment, redirecting airflow downward to maximize pressure differential even at a 0° angle of attack.
- **Low-Re Boundary Layer Control:** Engineered for micro-heavy lift competitions like SAE Aero Design, providing a balanced high-lift/low-drag envelope without the extreme low-Reynolds hysteresis of higher-camber sections.
- **Structural Depth:** Its 12.0% relative thickness provides deep internal spar volume, increasing bending moment resistance for heavy payload structures.

Aircraft B Selection: NACA 0012

- **Symmetrical Profile:** Identical upper and lower surfaces (0.0% camber) yield identical lift and stall characteristics in upright and inverted flight.
- **Zero Pitching Moment:** Generates a pitching moment coefficient (Cm) of exactly 0 at its aerodynamic center, preventing unwanted pitch changes during rapid roll maneuvers and banking transitions.
- **Low Minimum Drag:** Low baseline drag (Cd,min = 0.0054) at 0° angle of attack allows for clean momentum retention during dynamic aerobatic loops and turns.
- **Predictable Handling:** Provides a gentle, progressive trailing-edge stall without sudden lift drop during high-G transitions.

4. Quantitative Performance Comparison Matrix

Airfoil	CL,max	CD,min	Maximum L/D	Stall Characteristics	Max Camber (% chord)	Max Thickness (% chord)	Remarks
S1210 (Selected A)	1.91 (modelled)	0.0129 (modelled)	88 (modelled)	Designed for heavy lift; potentially soft stall	6.7%	12.0%	Balanced high-lift/low-drag profile; strong candidate for heavy-lift payload UAVs.
FX 63-137	1.66 (measured)	0.0100 (measured)	89 (modelled)	Good stall quality around Re 200k; complex behavior lower	5.94%	13.59%	High aerodynamic efficiency (L/D = 89); ideal choice for

Airfoil	CL,max	CD,min	Maximum L/D	Stall Characteristics	Max Camber (% chord)	Max Thickness (% chord)	Remarks
							gliders and long-endurance.
Clark-Y	1.40 (measured)	0.0100 (measured)	73 (modelled)	Conventional profile; practical, but not inherently stall-proof	3.55%	11.72%	Classic baseline reference airfoil; flat bottom surface simplifies construction.
S1223	2.22 (measured)	0.0180 (measured)	71 (modelled)	High lift; hysteresis and complex low-Re stall possible	8.5%	12.1%	High-lift low-Re design; monitor laminar separation bubble and hysteresis.
S9104	2.59 (predicted)	Not established	Not established	Predicted high lift; experimental stall evidence insufficient	14.8%	12.1%	Extreme camber profile; requires experimental/CFD validation for drag penalties.
S9104BTE	2.79 (predicted)	Not established	Not established	Highest predicted lift; blunt trailing edge adds drag	14.8%	12.0%	Highest CL,max variant; blunt trailing edge introduces

Airfoil	CL,max	CD,min	Maximum L/D	Stall Characteristics	Max Camber (% chord)	Max Thickness (% chord)	Remarks
							structural gain but base drag penalty.
NACA 0012 (Selected B)	1.40 (measured)	0.0054 (measured)	75 (modelled)	Gentle and progressive trailing-edge stall	0.0%	12.0%	Classic symmetric standard; zero pitching moment, widely used for aerobatics and rotorcraft.
NACA 0009	1.18 (measured)	0.0044 (measured)	69 (modelled)	Sharper leading-edge stall onset due to thin nose radius	0.0%	9.0%	Thin symmetric profile; low parasite drag, commonly used for tail surfaces.
S8064	1.25 (measured)	0.0090 (measured)	56 (modelled)	Smooth, progressive low-Re stall characteristic	1.1%	12.3%	Selig low-Re design for Q-500 RC pylon racing; optimized for low drag at moderate lift.
NACA 23012	1.76 (measured)	0.0060 (measured)	85 (modelled)	Relatively abrupt stall	1.84%	12.0%	High-lift 5-digit profile

Airfoil	CL,max	CD,min	Maximum L/D	Stall Characteristics	Max Camber (% chord)	Max Thickness (% chord)	Remarks
				without prominent pre-stall buffer			with near-zero pitching moment (Cm,c/4 ≈ 0).
NACA 2412	1.49 (measured)	0.0057 (measured)	94 (modelled)	Smooth, gentle trailing-edge stall with reliable warning	2.0%	12.0%	Standard 4-digit GA airfoil; well-balanced L/D and forgiving handling.

5. Influence of Airfoil Geometry on Aircraft Performance

- **Camber (Curvature of the Mean-Line):** Dictates lift generation and pitching moment. Higher camber accelerates upper-surface flow, raising maximum lift coefficient (CL,max) and producing lift at 0° angle of attack. Symmetrical airfoils (zero camber) produce zero lift at zero angle of attack. Higher camber also increases the nose-down pitching moment.
- **Thickness-to-Chord Ratio (t/c):** Thicker airfoils (~12–16%) provide internal structural depth for wing spars, increase maximum lift/drag ratios up to ~16% thickness, and exhibit gentler, more forgiving trailing-edge stall behavior. Thin airfoils (~8–10%) minimize parasite drag for rapid maneuvers, but offer less internal volume and sharper stall margins.
- **Maximum Thickness & Camber Position:** Moving maximum thickness or camber aft along the chord generally delays boundary layer turbulence transition, affecting profile drag and pitch stability.
- **Leading-Edge Radius & Trailing-Edge Shape:** A larger LE radius smooths out pressure peaks at high angles of attack, softening the stall boundary. Trailing-edge shape influences boundary layer separation and base drag penalties.

6. Single Airfoil Multi-Mission Feasibility & Trade-Off Analysis

No single fixed airfoil profile can satisfy both missions optimally without severe compromise.

- **Using S1210 for Aerobatics:** The 6.7% camber produces severe flow separation and high drag when flying inverted. Achieving negative lift requires highly negative angles of

attack, risking sudden inverted stall. Additionally, high nose-down pitching moment (Cm0) causes sluggish roll responses and heavy tail trim drag.

- **Using NACA 0012 for Heavy Payload:** Lower CL,max (1.40 vs 1.91) forces the aircraft to fly 16.8% faster to achieve takeoff lift for the same payload weight, requiring significantly longer field lengths and higher motor thrust output.
- **Dynamic Compromise:** Variable geometry flaps on a symmetrical profile (NACA 0012) can temporarily deploy camber for low-speed high-payload takeoff, then retract to 0° camber for neutral aerobatics.

7. Airfoil Characteristics vs. Whole-Aircraft Factors

- **Wing Loading (W/S) vs. Section CL:** While section CL,max limits total lift coefficient, Wing Loading (W/S) determines actual stall speed (V_stall) and turn radius. Payload designs pair high CL airfoils with generous wing areas, whereas aerobatic designs rely on low wing loading for rapid turn performance.
- **Aspect Ratio (AR) & Induced Drag:** 2D airfoil sections only account for profile drag (Cd0). 3D wings generate induced drag (CDi = CL^2 / (pi * AR * e)). High-payload aircraft require high aspect ratio wings (>10) to minimize induced drag during low-speed climb, whereas aerobatic wings use lower aspect ratios (4–6) to reduce roll inertia.
- **Control Authority & Stability:** Airfoil pitching moment (Cm0) dictates horizontal tail sizing and trim drag. A highly cambered profile requires larger tail volume coefficients (Vh) and trim download, whereas symmetrical airfoils eliminate trim changes during speed transitions.
- **Thrust-to-Weight Ratio (T/W) & Structure:** Airfoil thickness defines internal spar height for structural strength. High-payload designs pair thick profiles with high static thrust props, whereas aerobatic aircraft require T/W > 1.0 for continuous vertical maneuvers.